Name: ____________

Date: ____________

Exponents & Radicals

Student learning objectives for this unit

SUPPORTING SKILLS LEGEND

I — Introduce | D — Deepen | A — Apply | R — Reinforce

☐ **1. I can simplify expressions containing negative exponents.**

RELATED SKILLS

- I — Rewrite a negative exponent using a reciprocal.
- A — Apply the product rule for exponents.
- A — Apply the quotient rule for exponents.

☐ **2. I can rewrite expressions between radical and rational-exponent forms.**

RELATED SKILLS

- I — Convert between radical and rational-exponent notation.
- I — Interpret numerator and denominator roles in a rational exponent.

☐ **3. I can simplify radical expressions.**

RELATED SKILLS

- I — Extract perfect-power factors from a radical.
- A — Express a whole number as a product of prime factors.

☐ **4. I can apply exponent rules to expressions with integer and rational exponents.**

RELATED SKILLS

- I — Apply the zero-exponent rule.
- D — Apply the power rule for exponents.
- D — Apply the product rule for exponents.
- D — Apply the quotient rule for exponents.
- A — Interpret numerator and denominator roles in a rational exponent.
- A — Rewrite a negative exponent using a reciprocal.

☐ **5. I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.**

RELATED SKILLS

- I — Check a candidate solution for extraneous results.
- I — Isolate a radical expression before solving.
- I — Require an even-root radicand to be nonnegative when determining domain.
- A — Use inverse operations to maintain an equivalent equation.